Nibraas Khan

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Cambridge, MA - USA

SUMMARY

Background in applied AI (multi-agent LangGraph pipelines and NL-to-SQL for enterprise clients), industry research (reinforcement learning for context and memory management, VLA models for human-robot interaction), and academic research (CS PhD candidate, 25+ publications). Seeking applied AI and ML roles.

EDUCATION

- **Vanderbilt University** — *Aug 2020 - July 2026 Expected*
  Ph.D. Computer Science — Nashville, TN
  - **Dissertation:** *From Prediction to Explanation: Interpretable Systems for Human Motion Analysis in High-Impact Domains.*
- **Middle Tennessee State University** — *Aug 2017 - May 2020*
  B.S. Computer Science — Murfreesboro, TN

EXPERIENCE

- **Mitsubishi Electric Research Laboratories** — *Jan 2026 - Present*
  Research Scientist Intern — Cambridge, MA
  - Developed a PPO-based framework for adaptive memory and context management across Neural Processes and foundation models under bounded compute; 17% improvement over heuristic baselines.
  - Built pretraining and post-training recipes for on-device vision-language-action models enabling at-home human-robot interaction, fusing vision, proprioception, affective signals, and WiFi-CSI scene maps.
- **Actual Insight** — *Oct 2025 - Jan 2026*
  Applied Scientist Intern - Multi-Agent Systems — Nashville, TN
  - Designed and deployed multi-agent LLM pipelines (LangGraph) for 15+ enterprise clients: agent orchestration, output-to-source provenance, role-based access control, and PII redaction.
  - Built a LangChain natural-language-to-SQL workflow that let business users query their data in plain English, with a locality-sensitive-hashing cache that cut prompt-token usage by 18%.
  - Built a system that matches job candidates to roles via an embedding and pgvector similarity pipeline, with LLM scoring agents and an A/B testing harness to measure and improve match quality.
- **Vanderbilt University - Robotics and Autonomous Systems Lab** — *May 2020 - Present*
  Graduate Researcher — Nashville, TN
  - Pretrained a 30M-parameter transformer foundation model (PyTorch) with quaternion-based reconstruction on 15 IMU datasets (6 days on A100s) and fine-tuned for movement-quality assessment.
  - Engineered the full sensor-to-feedback stack for boxing: custom IMU/ESP32 hardware, firmware, and a React app with sub-second 3D-avatar feedback at 100 Hz; validated with 20 athletes, 6 coaches.

SELECTED PUBLICATIONS

- **Khan, N.**, Wichern, G., Laughman, C. R. 2026. *Reinforced Neural Processes: Memory-Efficient Time-Series Forecasting with a World-Feedback-Trained Memory Policy*. ICML 2026 RLxF Workshop (accepted).
- **Khan, N.**, Cole, K., Sarkar, N. 2026. *Toward Assessing Functional Decline in Mild Cognitive Impairment Using Wearable Sensors and Explainable Machine Learning*. Digital Health.

SKILLS

- **Foundation Models & LLMs:** Pretraining, Fine-Tuning, RAG, Embeddings, Agentic/Multi-Agent (LangGraph), VLMs/VLA
- **ML Methods:** Reinforcement Learning (PPO), Explainable AI, Few-Shot & Meta-Learning, Time-Series, Deep Learning (Transformers, RNNs, CNNs), Computer Vision
- **Systems & Hardware:** Embedded firmware (ESP32), IMU & multimodal sensor integration, real-time on-device inference, full-stack (React, TypeScript)
- **Frameworks & MLOps:** PyTorch, TensorFlow/Keras, Hugging Face Transformers, Scikit-learn, AWS (SageMaker, EC2, S3), Docker, Kafka, Airflow, Git
- **Languages:** Python, C#, Kotlin, TypeScript, SQL, Bash